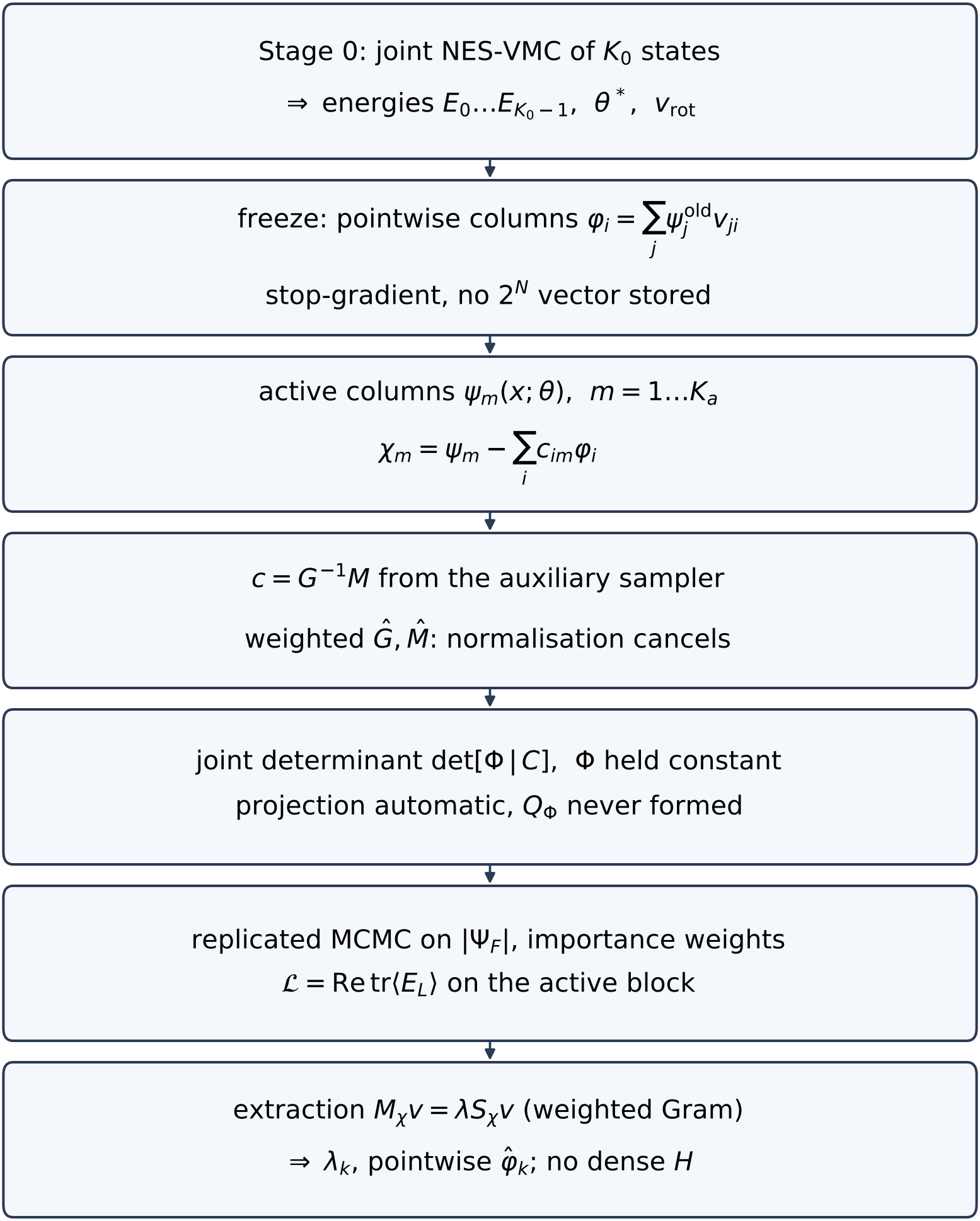
Stage 0: joint NES-VMC of K_0 states
$\Rightarrow$ energies $E_0 \ldots E_{K_0-1}$, θ^*, v_{rot}
freeze: pointwise columns $\varphi_i = \sum_j \psi_j^{\mathrm{old}} v_{ji}$
stop-gradient, no 2^N vector stored
active columns $\psi_m(x;\theta)$, $m = 1 \ldots K_a$
$\chi_m = \psi_m - \sum_i c_{im}\varphi_i$
$c = G^{-1}M$ from the auxiliary sampler
weighted $\hat{G}, \hat{M}$: normalisation cancels
joint determinant $\det[\Phi \,|\, C]$, Φ held constant
projection automatic, Q_Φ never formed
replicated MCMC on $|\Psi_F|$, importance weights
$\mathcal{L} = \mathrm{Re}\,\mathrm{tr}\langle E_L \rangle$ on the active block
extraction $M_\chi v = \lambda S_\chi v$ (weighted Gram)
$\Rightarrow$ λ_k, pointwise $\hat{\varphi}_k$; no dense H